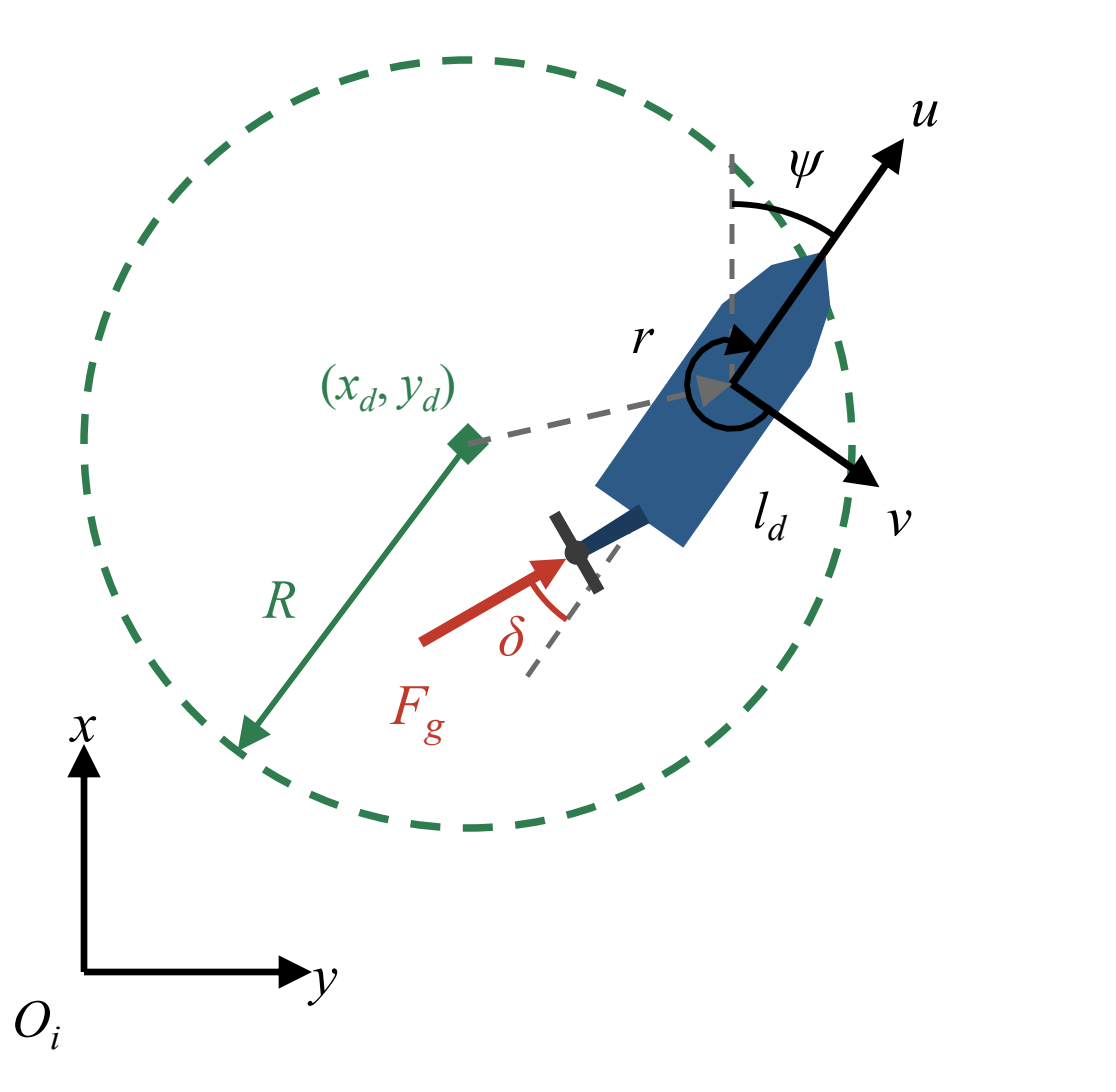
u
ψ
r
$(x_d, y_d)$
$l_d$
v
R
δ
$F_g$
x
y
$O_i$